

AUTISM

Epidemiologic Evidence

The committee reviewed one study to evaluate the risk of autism after the administration of DTaP vaccine. This one study (Geier and Geier, 2004) was not considered in the weight of epidemiologic evidence because it provided data from a passive surveillance system and lacked an unvaccinated comparison population.

Weight of Epidemiologic Evidence

The epidemiologic evidence is insufficient or absent to assess an association between diphtheria toxoid-, tetanus toxoid-, or acellular pertussis-containing vaccine and autism.

Mechanistic Evidence

The committee did not identify literature reporting clinical, diagnostic, or experimental evidence of autism after the administration of vaccines containing diphtheria toxoid, tetanus toxoid, and acellular pertussis antigens alone or in combination.

Weight of Mechanistic Evidence

The committee assesses the mechanistic evidence regarding an association between diphtheria toxoid-, tetanus toxoid-, or acellular pertussis-containing vaccine and autism as lacking.

Causality Conclusion

Conclusion 10.6: The evidence is inadequate to accept or reject a causal relationship between diphtheria toxoid-, tetanus toxoid-, or acellular pertussis-containing vaccine and autism.⁷⁸

It is troubling that the only study the IOM could locate regarding whether DTaP causes autism, (Geier and Geier, 2004), concluded there *was* an association between DTaP and autism.⁷⁹ No research has been published since 2011 that could change the IOM's conclusion. Based on the foregoing, the CDC cannot validly make the blanket assertion that "Vaccines Do Not Cause Autism."

As with DTaP, there are also no published studies showing that autism is not caused by Hepatitis B, Rotavirus, Hib, Pneumococcal, Inactivated Poliovirus, Influenza, Varicella, or Hepatitis A vaccines – all of which HHS recommends babies receive by one year of age.⁸⁰

Instead, HHS's claim that "Vaccines Do Not Cause Autism" relies almost entirely upon studies exclusively studying only one vaccine, MMR (which is administered no earlier than one year of age), or only one vaccine ingredient, thimerosal, with regard to autism.⁸¹ Putting aside the controversy surrounding these studies, studies which focus on only one vaccine and one ingredient while ignoring the entire balance of the CDC's pediatric vaccine schedule cannot support the

⁷⁸ Ibid.

⁸⁰ <https://www.cdc.gov/vaccines/schedules/hcp/imz/child->

*found that evidence assessing outcomes in sub populations of children who may be potentially susceptible to adverse reactions to vaccines (such as children with a family history of autoimmune disease or allergies or children born prematurely) was limited and is characterized by uncertainty about the definition of populations of interest and definitions of exposures and outcomes.*⁶⁹

HHS had failed to even define the terminology for the study of susceptible subpopulations; hence IOM admonished HHS to “develop a framework that clarifies and standardizes definitions of ... populations that are potentially susceptible to adverse events.”⁷⁰ While every vaccine brand is the same, it is plain that every child is different.

The IOM correctly points out in 2011 that given the “widespread use of vaccines” and “state mandates requiring vaccination of children ... it is essential that safety concerns receive assiduous attention.”⁷¹ This is the same call for diligent attention that the IOM made in 1991 and 1994. Unfortunately, all of these calls for action have gone unheeded. The critical scientific inquiry to identify individuals susceptible to serious vaccine injury has simply never commenced.

Since the IOM’s first call for this science in 1991, HHS has spent tens of billions promoting and purchasing vaccines, and

vaccine makers have accumulated hundreds of billions in vaccine revenue.⁷² Yet, during this time, no material funds have been allocated to identify susceptible subpopulations, let alone what injuries are caused by vaccines.⁷³

4. CDC Views Vaccine Safety as a Public Relations Issue

The CDC, unfortunately, has treated vaccine safety as a public relations issue rather than a public health imperative. For example, the CDC claims on its website that “Vaccines Do Not Cause Autism” even though this broad claim is plainly not supported by the scientific literature.⁷⁴

Indeed, as part of the IOM’s 2011 review of vaccine safety, it was asked by HHS whether there is a causal relationship between autism and the DTaP vaccine administered to children at two, four, six, and fifteen months of age.⁷⁵ The IOM could not locate a single study supporting that DTaP does not cause autism.⁷⁶ The IOM therefore concluded: “The evidence is inadequate to accept or reject a causal relationship between diphtheria toxoid–, tetanus toxoid–, or acellular pertussis–containing vaccine and autism.”⁷⁷ The IOM’s full explanation for this finding is as follows:

⁶⁹ <https://www.nap.edu/read/13563/chapter/9#130>

⁷⁰ Ibid.

⁷¹ <https://www.nap.edu/read/13164/chapter/3#28>

⁷² <https://www.hhs.gov/about/budget/index.html#previous>

\$5 billion promoting and purchasing vaccines (Ibid.), the CDC Immunization Safety Office’s budget is apparently only around \$20 million. [http://www.ajpmonline.org/article/S0749-3797\(15\)00314-1/pdf](http://www.ajpmonline.org/article/S0749-3797(15)00314-1/pdf)

A. Autism Is Growing at Epidemic Proportions

1. Introduction

Autism was once considered a rare disease that affected an estimated 1 in 10,000 individuals in the United States. The Committee held its first hearing on the dramatic rise in autism in April of 2000. At the time, Federal agencies were estimating that autism affected 1 in 500 children in the United States. By 2002, the National Institutes of Health had adjusted that rate to 1 in 250 children in the United States. The Autism Society of America estimates that the number of autistic children is growing by 10 to 17 percent each year.

In that first hearing, Chairman Burton reported that according to U.S. Department of Education statistics, requests for services for school-age children with autism spectrum disorders had risen dramatically in every state.

Mr. Burton: ``California has reported a 273 percent increase in children with autism since 1988 . . . Florida has reported a 571 percent increase in autism. Maryland has reported a 513 percent increase between 1993 and 1998 . . . In 1999, there were 2,462 children ages 3 to 21 in Indiana diagnosed with autism. That is one-fourth of 1 percent of all the school children in Indiana, or 1 out of every 400 . . . This increase is not just better counting. If we want to find a cure, we must first look to the cause.''

In July 2000, Dr. Stephanie Cave shared her observations about the rapid growth of autism and the pressures it is placing on families and medical professionals:

``I am in family practice in Baton Rouge, LA. I want to express my deep appreciation to you and to the members of the committee for allowing me to testify. I am presently treating over 300 autistic children, with an additional 150 waiting to get in.

``We are treating children from all over the United States and getting calls from many places around the globe. This is truly an epidemic. If you have any idea that it is not, I invite you to sit in my office for 2 hours.''

2. Studies Are Documenting the Incredible Growth of Autism

In the 1990's, the CDC conducted two prevalence studies that confirmed dramatic spikes in autism cases. One was conducted in Brick Township, New Jersey, the other in Atlanta, Georgia.

In late 1997, after noticing an apparently larger than expected number of children with autism in their community, a citizen's group in Brick Township, New Jersey, contacted the New Jersey Department of Health and Senior Services (DHSS). Because of the complexity of the disorder and the concerns that environmental factors might play a role, the New Jersey DHSS, U.S. Senator Robert Torricelli, and U.S. Representative Christopher Smith contacted the CDC and the ATSDR for assistance. In response, the CDC

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conducted an extensive prevalence investigation.

The rate of autism among children in Brick Township was 4 per 1,000 (1 in 250) children aged 3 through 10 years. The prevalence of the more broadly defined autism spectrum disorder was 6.7 per 1,000 (1 in 150) children. It is important to note that even though the families of Brick Township requested that the CDC include an evaluation of a possible link between autism and their children's immunization, the CDC chose not to do so. Their evaluation of the cause of the cluster of autism in Brick Township was inconclusive.

The CDC's Atlanta study confirmed the dramatic results of the Brick Township study. The CDC found that 1,987 of the 289,456 children aged 3 to 10 years in metropolitan Atlanta in 1996 were autistic (1 in 146). These numbers were 10 times higher than studies conducted in the 1980s and early 1990s.

Last November, a study on autism in California determined that the number of autistic individuals in that state has nearly tripled. Equally important, the study stated that the increase was real, and could not be explained by changes in diagnostic criteria or better diagnoses. The study, funded by the state legislature and conducted by the University of California at Davis, determined that the number of autistic people in that state grew by 273% between 1987 and 1998.

The main author of the study, Dr. Robert Byrd, said, ``It is astounding to see a three-fold increase in autism with no explanation . . . there's a number of things that need to be answered. We need to rethink the causes of autism.''

The 2002 report confirmed a 210 percent increase in the number of new children professionally diagnosed with the most severe cases of autism entering the developmental services system between 2001 and 2002. The system added 3,577 new cases in 2002.

It is important to note that the figures reported in California do not include persons with Pervasive Developmental Disorder (PDD), PDD-Not Otherwise Specified

12. The CDC's failure to state a preference for thimerosal-free vaccines in 2000 and again in 2001 was an abdication of their responsibility. As a result, many children received vaccines containing thimerosal when thimerosal-free alternatives were available.

13. The Influenza vaccine appears to be the sole remaining vaccine given to children in the United States on a regular basis that contains thimerosal. Two formulations recommended for children six months of age or older continue to contain trace amounts of thimerosal. Thimerosal should be removed from these vaccines. No amount of mercury is appropriate in any childhood vaccine.

14. The CDC in general and the National Immunization Program in particular are conflicted in their duties to monitor the safety of vaccines, while also charged with the responsibility of purchasing vaccines for resale as well as promoting increased immunization rates.

15. There is inadequate research regarding ethylmercury neurotoxicity and nephrotoxicity.

16. There is inadequate research regarding the relationship between autism and the use of mercury-containing vaccines.

17. To date, studies conducted or funded by the CDC that purportedly dispute any correlation between autism and vaccine injury have been of poor design, under-powered, and fatally flawed. The CDC's rush to support and promote such research is reflective of a philosophical conflict in looking fairly at emerging theories and clinical data related to adverse reactions from vaccinations.

7. A growing number of scientists and researchers believe that a relationship between the increase in neurodevelopmental disorders of autism, attention deficit hyperactive disorder, and speech or language delay, and the increased use of thimerosal in

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vaccines is plausible and deserves more scrutiny. In 2001, the Institute of Medicine determined that such a relationship is biologically plausible, but that not enough evidence exists to support or reject this hypothesis.

8. The FDA acted too slowly to remove ethylmercury from over-the-counter products like topical ointments and skin creams. Although an advisory committee determined that ethylmercury was unsafe in these products in 1980, a rule requiring its removal was not finalized until 1998.

9. The FDA and the CDC failed in their duty to be vigilant as new vaccines containing thimerosal were approved and added to the immunization schedule. When the Hepatitis B and Haemophilus Influenzae Type b vaccines were added to the recommended schedule of childhood immunizations, the cumulative amount of ethylmercury to which children were exposed nearly tripled.

10. The amount of ethylmercury to which children were exposed through vaccines prior to the 1999 announcement exceeded two safety thresholds established by the Federal government for a closely related substance--methylmercury. While the Federal Government has established no safety threshold for ethylmercury, experts agree that the methylmercury guidelines are a good substitute. Federal health officials have conceded that the amount of thimerosal in vaccines exceeded the EPA threshold of 0.1 micrograms per kilogram of bodyweight. In fact, the amount of mercury in one dose of DTaP or Hepatitis B vaccines (25 micrograms each) exceeded this threshold many times over. Federal health officials have not conceded that this amount of thimerosal in vaccines exceeded the FDA's more relaxed threshold of 0.4 micrograms per kilogram of body weight. In most cases, however, it clearly did.

11. The actions taken by the HHS to remove thimerosal from vaccines in 1999 were not sufficiently aggressive. As a result, thimerosal remained in some vaccines for an additional two years.

II. Findings and Recommendations

A. Findings

Through this investigation of pediatric vaccine safety, the following findings are made:

1. Mercury is hazardous to humans. Its use in medicinal products is undesirable, unnecessary and should be minimized or eliminated entirely.

2. For decades, ethylmercury was used extensively in medical products ranging from vaccines to topical ointments as preservative and an anti-bacteriological agent.

3. Manufacturers of vaccines and thimerosal, (an ethylmercury compound used in vaccines), have never conducted adequate testing on the safety of thimerosal. The FDA has never required manufacturers to conduct adequate safety testing on thimerosal and ethylmercury compounds.

4. Studies and papers documenting the hyperallergenicity and toxicity of thimerosal (ethylmercury) have existed for decades.

5. Autism in the United States has grown at epidemic proportions during the last decade. By some estimates the number of autistic children in the United States is growing between 10 and 17 percent per year. The medical community has been unable to determine the underlying cause(s) of this explosive growth.

6. At the same time that the incidence of autism was growing, the number of childhood vaccines containing thimerosal was growing, increasing the amount of ethylmercury to which infants were exposed threefold.

7. A growing number of scientists and researchers believe that a relationship between the increase in neurodevelopmental disorders of autism, attention deficit hyperactive disorder, and speech or language delay, and the increased use of thimerosal in

[Congressional Record (Bound Edition), Volume 151 (2005), Part 10]
[House]
[Pages 13935-13938]
[From the U.S. Government Publishing Office, www.gpo.gov]

MERCURY AND AUTISM

The SPEAKER pro tempore. Under a previous order of the House, the gentleman from Indiana (Mr. Burton) is recognized for 5 minutes.

Mr. BURTON of Indiana. Mr. Speaker, I have been down here a lot talking about autism over the years and my committee had many hearings on the issue of autism. My grandson became autistic after receiving 9 shots in one day, 7 of which contained mercury, in a product called thimerosal. And he is doing better but it has been a very difficult time for me and my family.

I strongly believe that there is a link between the mercury that is in the thimerosal in the vaccines and children developing neurological disorders such as autism. In fact, according to a recent study released by collaboration of U.S. medical researchers from Johns Hopkins University, Northeastern University in Boston, and the University of Nebraska and Tufts University that was published in the Vancouver Sun in February of last year and was officially released in the April 2004 edition of the scientific journal Molecular Psychiatry, ``A recent review of vaccine-related adverse events in the U.S. found a significant correlation between shots containing thimerosal,' i.e. mercury ``and autism.''

The study further concluded that the use of thimerosal-containing shots could account for the rising rates of autism since the early 1980s when more thimerosal-containing vaccinations were added to the government-mandated childhood vaccination schedule.

Scientific evidence aside, we have seen an increase from 1 in 10,000 children who are autistic to 1 in 166 since they started using thimerosal in many, many vaccines in the early eighties and children started getting more of these shots.

I am not against vaccinations but I do believe, as many of my colleagues, including the gentleman from Florida (Mr. Weldon) believe, that mercury should be taken out of all childhood vaccines and in fact all vaccines.

Expert Opinion

1. Introduction
2. The autistic enterocolitis hypothesis
3. Epidemiological studies of MMR and autism
4. Other evidence related to MMR and autism
5. Causality assessment
6. Remaining questions
7. Conclusion and expert opinion

General

MMR vaccination and autism: is there a link?

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In 1998, a report was published describing 12 patients with inflammatory bowel conditions and regressive developmental disorders consisting primarily of autism. The authors hypothesised that MMR vaccine may have been responsible for the bowel dysfunction which subsequently resulted in the neurodevelopmental disorders. The suggestion that measles vaccine may cause autism through a persistent bowel infection generated much interest since it provided a possible biological mechanism for a causal association. Epidemiological studies, however, have not found an association between MMR vaccination and autism. Autism has a strong genetic component and its associated neurological defects probably occur during embryonic development. It seems unlikely that a vaccination that is given after birth could cause autism. In a minority of cases, autism may have onset after 1 year of age (regressive autism) but the one epidemiological study that included such cases did not find an association with MMR vaccination. Currently, the weight of the available epidemiological and related evidence does not support a causal link between MMR vaccine and autism.

Keywords: autism, epidemiology, inflammatory bowel disease, measles virus, MMR vaccine

Expert Opin. Drug Saf. (2002) 1(2):115-120

1. Introduction

It has been suggested that MMR vaccine may cause autism. The main evidence in support of such a possibility has been that in some children with autism, developmental delays or autistic behaviours become apparent within a few weeks to a few months after vaccination. The apparent onset in close proximity to vaccination, however, may be a coincidental temporal association. Typically, autism is first diagnosed or suspected during the second year of life, which is also when MMR vaccine is given (e.g., recommended at 12 – 15 months of age in the US).

The incidence of each of eleven chronic childhood conditions in vaccinated and unvaccinated children expressed as actual incidence and as a percentage of the total vaccinated and unvaccinated children surveyed.

	VACCINATED CHILDREN (n=226)		UNVACCINATED CHILDREN (n=269)	
	Numbers of children with condition	% of total vaccinated children	Numbers of children with condition	% of total unvaccinated children
Asthma	34	15.04	8	2.97
Eczema	63	27.88	34	12.64
Ear Infection/ Glue Ear	56	24.78	16	5.95
Grommets	14	6.19	2	0.74
Tonsillitis	26	11.50	3	1.12
Tonsillectomy	12	5.31	0	0.00
Apnoea	14	6.19	4	1.49
Hyperactivity	13	5.75	4	1.49
Diabetes	0	0.00	0	0.00
Epilepsy	4	1.77	0	0.00
Slow Development of Motor Skills	6	2.65	4	1.49

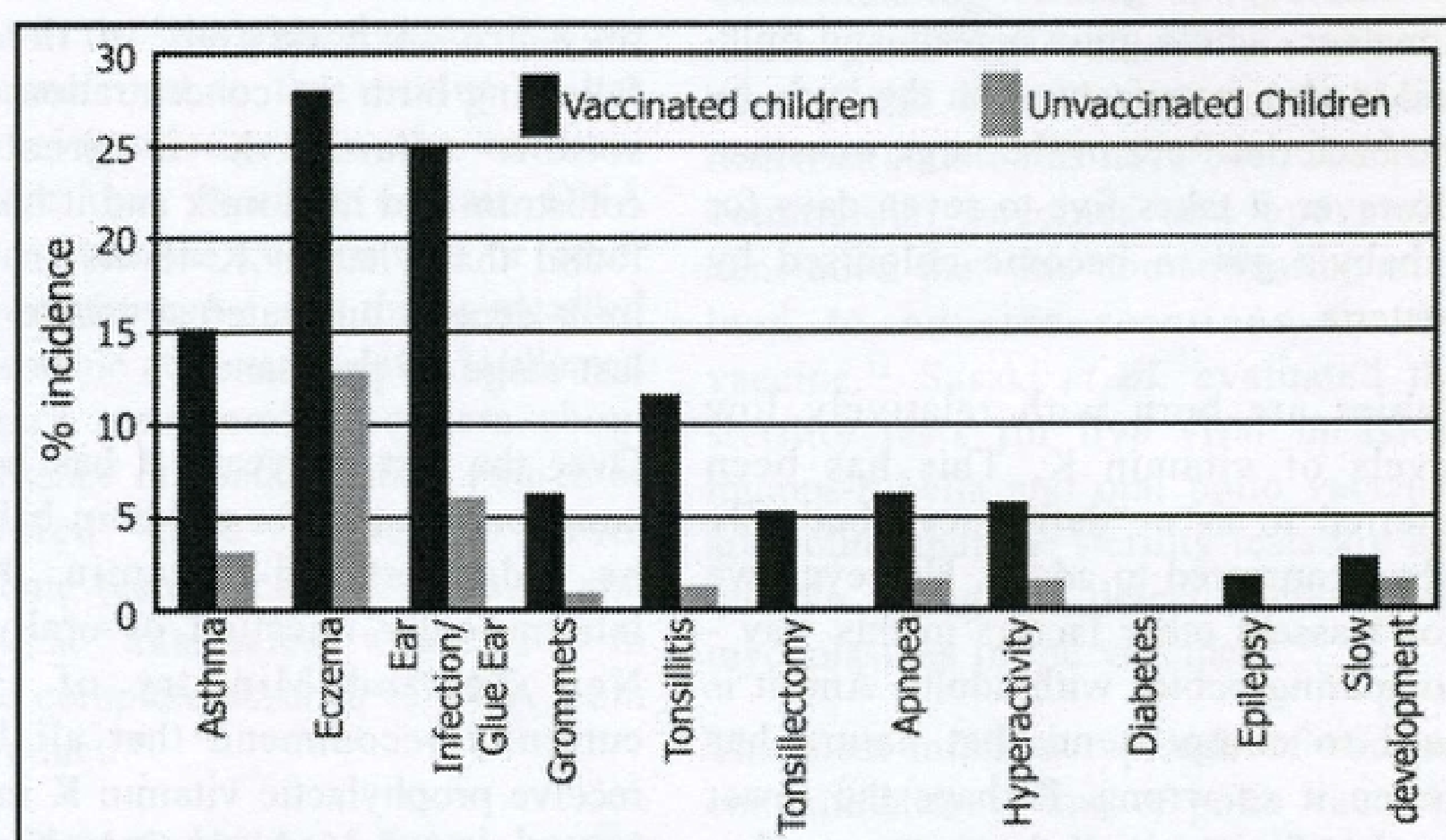
of vaccinated and 65% of unvaccinated children were still being breastfed - the national average is 39%; and over twelve months of age 27% of vaccinated and 35% of unvaccinated were still getting breastmilk while the national average had plummeted to just 6% (Plunket Society, Pers. comm., June 2001).

In an analysis of the role that breastfeeding and age-at-weaning played in the incidence of chronic childhood conditions in both the vaccinated and unvaccinated children surveyed, the data showed that there was no breastfeeding factor that could account for the difference in incidence between the two groups.

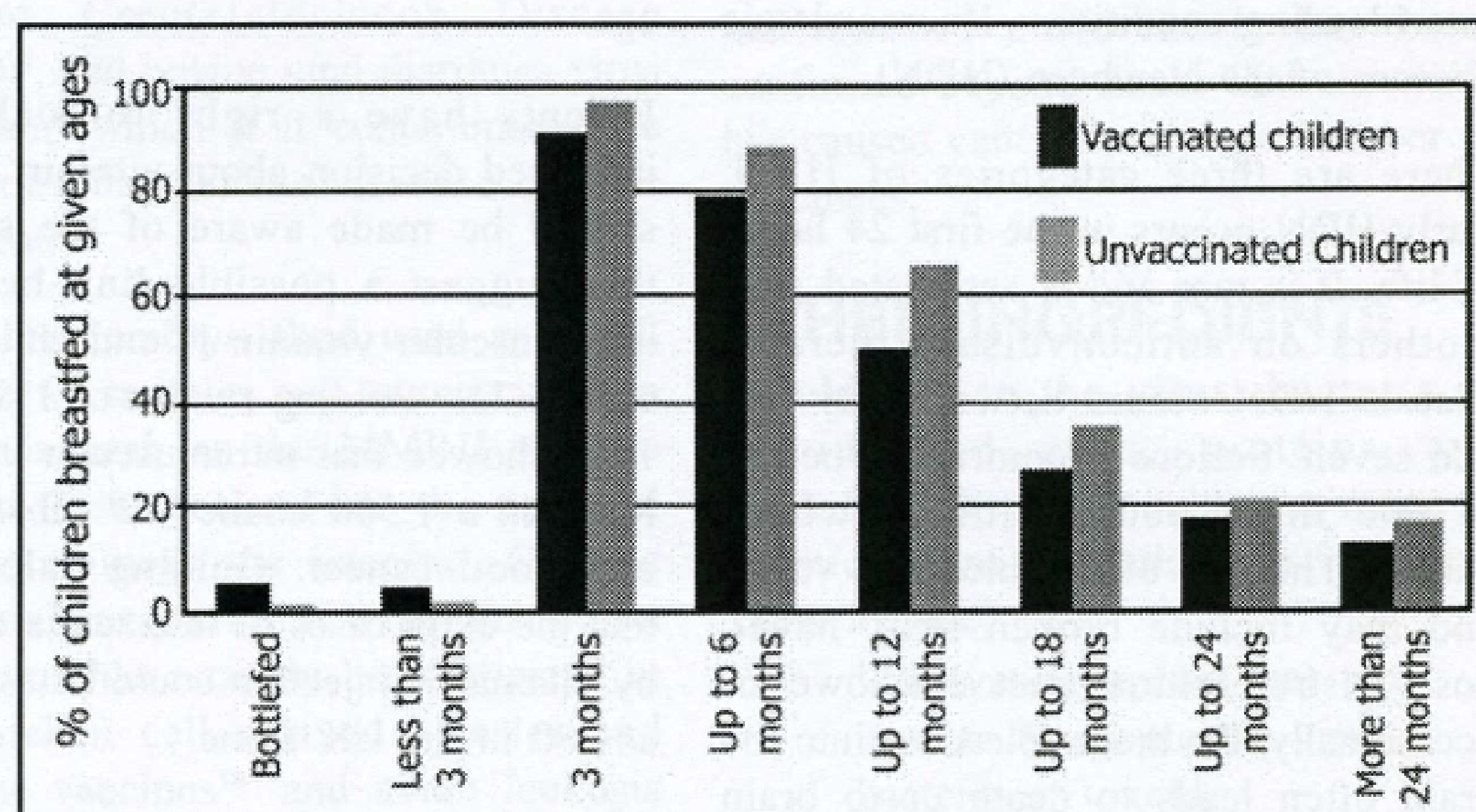
In other research, a study of 1265 Christchurch children born in 1977 found that none of the unvaccinated children had asthma or had had doctors consultations for asthma or allergic conditions.

"The 23 children who received no diphtheria/pertussis/tetanus (DPT) and polio immunizations had no recorded asthma episodes or consultations for asthma or other allergic illness before age 10 years; in the immunized children, 23.1% had asthma episodes, 22.5% asthma consultations, and 30.0% consultations for other allergic illness. Similar differences were observed at ages 5 and 16 years."¹

While this was a very limited study, particularly in terms of the numbers of unvaccinated children that were involved and the range of chronic conditions investigated, it provides solid scientific evidence in support of considerable anecdotal evidence that unvaccinated children are healthier than their vaccinated peers.



A graph of the incidence of each of eleven chronic childhood conditions in vaccinated and unvaccinated children expressed as a percentage of the total vaccinated and unvaccinated children surveyed.



The graph shows the percentage of surveyed children that were bottle-fed, breastfed for less than three months, and that were still being breastfed at 3 months, and up to 6, 12, 18 and 24 months in both the vaccinated and unvaccinated groups.

1 Kemp T, Pearce N, Fitzharris P, Crane J, Fergusson D, St George I, Wickens K, Beasley R.: Is infant immunization a risk factor for childhood asthma or allergy? *Epidemiology*. 1997 Nov; 8(6): 678-80.

Autism on rise; some cite vaccines

AMA doubts immunization link

By Chris Sewell

The Capital Times/Medill News Service

WASHINGTON — Autism is on the rise, and lawmakers are getting involved in the effort to find out why.

Autism is a developmental disorder that can be marked by symptoms ranging from varying repetitive behaviors to more severe conditions that can include mental retardation, epilepsy, poor sleeping habits and gastrointestinal disorders.



The U.S. Department of Education reports that autism increased by 154 per-

AUTISM, VACCINES, AND IMMUNE REACTIONS

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1. Introduction

Autism is the fastest-growing developmental disability in children today. Millions are affected worldwide, and the numbers are rising sharply to epidemic levels. Autism causes neurological and behavioral deficits in children, impairing their ability for social interaction, language, communication, cognition, and imagination. Autism is an idiopathic disorder of unknown etiology. Current theories include genetic factors, immune factors, viral factors, neural factors, and yet other unidentified factors. For last 10-12 years, I focused on autoimmune mechanism of pathogenesis and autoimmune therapy for patients affected with the disorder. I started out working on autism with the idea of "Autism as an Autoimmune Disorder" and my hope was to use autoimmunity as a target of drug development for treating autism [1-3]. Indeed, this is happening today [4-6]. In my presentation today, I will be speaking on the topic of "autism, vaccines, and immune reactions" and to examine the possibility of a causal link between vaccines and autism.

I view autism as a very complex disorder with different subsets and one such subset might be autoimmune in origin. I have studied autism as an autoimmune disorder, in which viral-autoimmune interplay may lead to pathological changes in the central nervous system (CNS). The essence of my "Autoimmune Hypothesis" is that a virus-induced autoimmune response to developing brain myelin may impair anatomical development of neural pathways in autistic children [3,4]. This line of thinking relies on the fact that the speed of nerve-impulse transmission depends essentially on structural properties of the insulating myelin sheath, connecting nerve fibers, and axon diameter. The anatomical changes could ultimately lead to life-long disturbances of higher mental functions such as learning, memory, communication, social interaction, etc. Fundamentally, therefore, I think that autism can be treated successfully using some of the therapies proven effective in treating other autoimmune diseases. To that end, however, the complete identification and characterization of autoimmune pathology in autism is of utmost importance today.

Autoimmune Hypothesis

Environmental Factors (viruses/vaccines) → Immune Dysfunction →

Autoimmunity to Brain → Autism

2. Viruses as causal factors in autism: *Measles is likely an etiological agent.*

Leading scientists in the field commonly believe that the viral infections trigger autoimmune responses and eventually lead to organ-specific autoimmune diseases. In autism, the trigger mechanism is not known but viral infections have been suspected. Viruses can enter the brain through the nasopharyngeal membranes or induce an autoimmune response against the brain, thereby impact the development of the central nervous system (CNS). Since the onset of the disorder is quite early on in life, viruses might serve as teratogens (agents that cause developmental malfunctions) etiologically linked to autism. Children with congenital rubella had certain autistic-like behaviors. Some autistic children did not make antibodies to rubella vaccine even after the repeated rubella immunization. Few cases of autism have also been described among children with congenital cytomegalovirus (CMV) infection.

Recently, we took a novel approach of studying viral etiology in autism [7-9]. We raised two simple questions: First, do autistic children harbor abnormal virus serology (antibody levels) and, secondly, is there a correlation between virus serology and brain antibodies. We studied immune response to viruses by measuring the level of their antibodies. For this purpose, we measured antibodies to five viruses: Measles virus, mumps virus, rubella virus, CMV, and human herpesvirus-6 (HHV-6). To our surprise, we found that the antibody level of only the measles virus, but not of the other viruses tested, was significantly higher in autistic children than the normal children [reference #4 and #9; Table 1 and Fig. 1]. In addition, we found an interesting correlation between measles antibody and brain autoimmunity, which was marked by myelin basic protein (MBP) antibodies. These two immune markers correlated in 90% or greater autistic children, suggesting a causal link of measles virus with autoimmunity in autism. But the serology to other viruses and other brain autoantibodies did not show this correlation. This was a very important finding that prompted us to postulate a temporal link of measles virus in the etiology of the disorder [Singh et al., 1998; reference #7]. To that end, it is also noteworthy that the immune manifestations of measles virus infection are quite similar to the immune abnormalities in autistic children, indirectly pointing to an etiological link of measles infection in autism.

Table 1. Viral antibodies in the sera of autistic children

Virus antibody	Subject group	EIA Units (mean $\pm$ SD)	p value
CMV-IgG	Autistic (n=30)	0.23 $\pm$ 0.32	0.37
	Normal (n=30)	0.28 $\pm$ 0.46	
HHV-6-IgG	Autistic (n=45)	2.18 $\pm$ 5.35	0.459
	Normal (n=37)	1.52 $\pm$ 0.64	
Rubella-IgG	Autistic (n=31)	3.59 $\pm$ 1.19	0.076
	Normal (n=12)	2.90 $\pm$ 0.81	
Mumps-IgG	Autistic (n=30)	2.57 $\pm$ 1.50	0.759
	Normal (n=32)	2.46 $\pm$ 1.31	
Measles-IgG	Autistic (n=42)	3.83 $\pm$ 1.23	0.004 (Significant)
	Normal (n=26)	3.08 $\pm$ 0.45	

THE INSTITUTE OF MEDICINE (IOM) MEETING ON VACCINES AND AUTISM
Washington, DC, USA (February 9, 2004); Presented by: *Vijendra Singh, PhD*

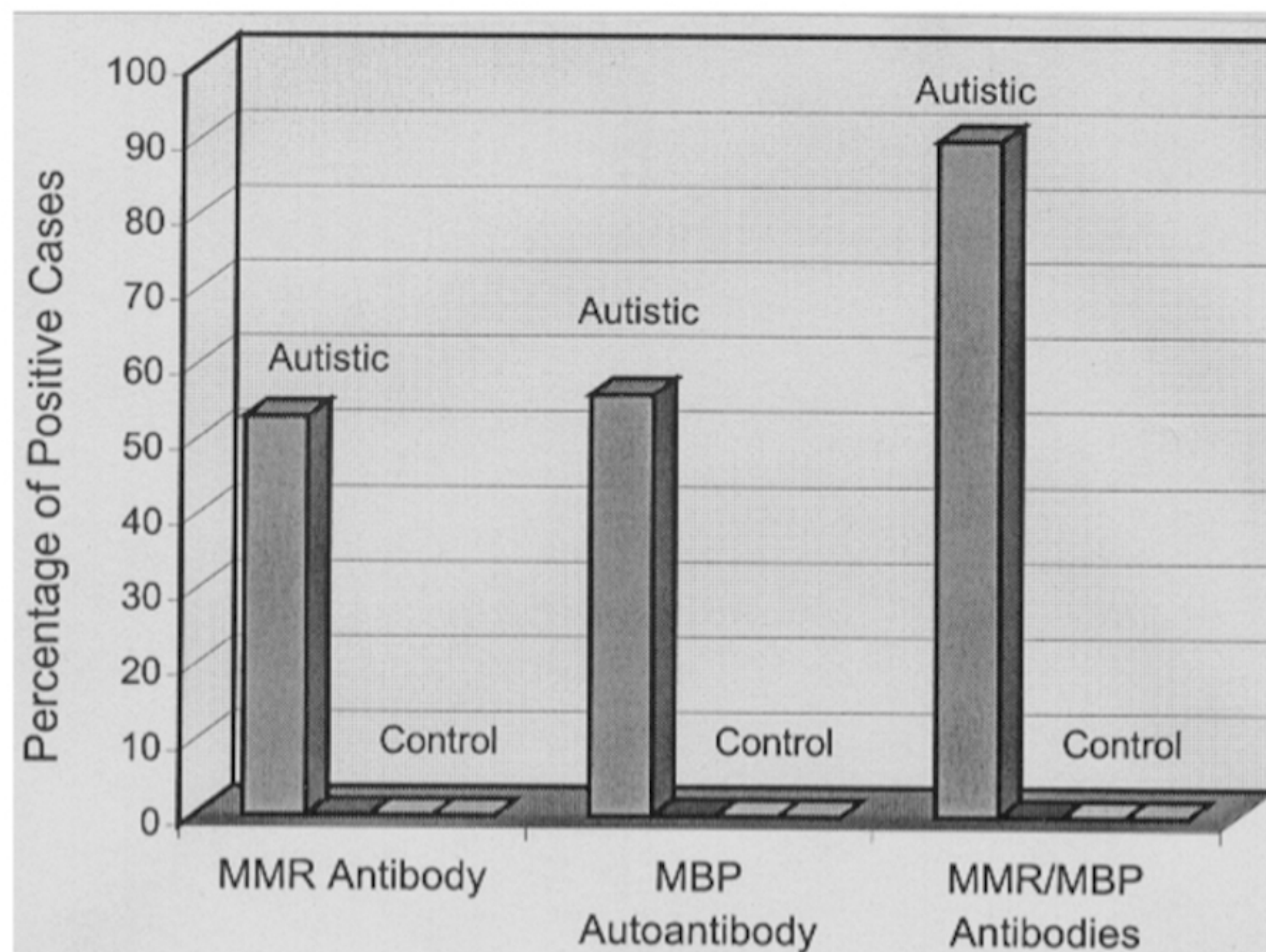


Figure 3. MMR and MBP antibodies in autistic children. **Note that 90% or more correlation was found between MMR and MBP antibodies in autistic children** (vertical bars) but not in controls that included normal children (baseline box 1), normal siblings (baseline box 2), and other disease children (baseline box 3).

4. Immune reactions in autism: Autoimmunity is the core of the problem.

Several groups have demonstrated abnormal immune reactions, in particular autoimmunity, in children with autism and related spectrum disorders. Autoimmunity is an abnormal immune reaction in which the immune system goes haywire and reacts abnormally against body's own organs, and the end result is an autoimmune disease. Several factors contribute to autoimmune diseases. Microbes such as viruses can trigger autoimmune diseases. They are generally linked to certain genes that control immune responses. They

6. Immune Modulation Therapy (IMT) in Autism

Accumulating evidence suggest that autoimmunity plays a key role in the pathogenesis of autism. The idea that autism is an autoimmune disorder is further strengthened by the fact that autistic patients respond well to treatment with immune modulating drugs [4-6]. Immune interventions can produce immune modulation – a state of suppression or stimulation. Since autistic patients do not show a classical primary immunodeficiency, simply boosting their immunity is not a good strategy. However, they do have immune abnormalities and therefore, depending on the nature of the immune abnormality, the goal of IMT should be to normalize or reconstitute the immune function. This will permit a more balanced immune response, avoiding major fluctuations of overt immune activity that could be detrimental to the patient. The IMT should always be given in consultation with a physician, preferably a clinical immunologist, allergist or hematologist. For autism, the recommended list of IMT includes steroid therapy, immunoglobulin therapy, oral autoantigen therapy, transfer factor therapy, immunomodulating drugs, and plasmapheresis.

7. Summary and concluding remarks

1. Considering an estimated population of 500,000 Americans with autism (not including all spectrum disorders), I think that between 250,000 to 350,000 autistic patients could benefit directly from autoimmunity research today.
2. The current genetic research estimates that no more than 10% of all autistic cases are genetic in origin. Simply put, the remainder 90% of autistic cases is sporadic with a non-genetic etiology. I tend to think that the sporadic form is by and large an “acquired” subset involving autoimmunity. This subset is likely triggered by a virus, possibly measles virus or MMR vaccine. I recently designated this subset as an “Autoimmune Autistic Disorder (AAD)” – a term coined to describe autoimmune subset of autism (Singh VK: May 1, 2003). I think that the autoimmunity research has a global impact for treating autism worldwide hence the physicians and researchers should pay a closer attention to autoimmunity research in autism.

CBS NEWS INVESTIGATES

Family to Receive \$1.5M+ in First-Ever Vaccine-Autism Court Award



By Sharyl Attkisson
September 10, 2010 / 10:44 AM EDT / CBS News



The first court award in a vaccine-autism claim is a big one. CBS News has learned the family of Hannah Poling will receive more than \$1.5 million dollars for her life care; lost earnings; and pain and suffering for the first year alone.

In addition to the first year, the family will receive more than \$500,000 per year to pay for Hannah's care. Those familiar with the case believe the compensation could easily amount to \$20 million over the child's lifetime.



Nine-year-old Hannah Poling.

AP PHOTO/ATLANTA JOURNAL-CONSTITUTION, JOHN SPINK

Hannah was described as normal, happy and precocious in her first 18 months.

Then, in July 2000, she was vaccinated against nine diseases in one doctor's visit: measles, mumps, rubella, polio, varicella, diphtheria, pertussis, tetanus, and Haemophilus influenzae.

Afterward, her health declined rapidly. She developed high fevers, stopped eating, didn't respond when spoken to, began showing signs of autism, and began having screaming fits. In 2002, Hannah's parents filed an autism claim in federal vaccine court. Five years later, the government settled the case before trial and had it sealed. It's taken more than two years for both sides to agree on how much Hannah will be compensated for her injuries.

Epidemiologic and Molecular Relationship Between Vaccine Manufacture and Autism Spectrum Disorder Prevalence

[Theresa A Deisher](#), [Ngoc V Doan](#), [Kumiko Koyama](#), [Sarah Bwabye](#)

PMID: 26103708

Abstract

Objectives: To assess the public health consequences of fetal cell line manufactured vaccines that contain residual human fetal DNA fragments utilizing laboratory and ecological approaches including statistics, molecular biology and genomics.

Method: MMR coverage and autism disorder or autism spectrum disorder prevalence data for Norway, Sweden and the UK were obtained from public and government websites as well as peer reviewed published articles. Biologically, the size and quantity of the contaminating fetal DNA in Meruvax II and Havrix as well as the propensity of various cell lines for cellular and nuclear uptake of primitive human DNA fragments were measured and quantified using gel electrophoresis, fluorescence microscopy and fluorometry. Lastly, genomic analysis identified the specific sites where fetal DNA fragment integration into a child's genome is most likely to occur.

Results: The average MMR coverage for the three countries fell below 90% after Dr. Wakefield's infamous 1998 publication but started to recover slowly after 2001 until reaching over 90% coverage again by 2004. During the same time period, the average autism spectrum disorder prevalence in the United Kingdom, Norway and Sweden dropped substantially after birth year 1998 and gradually increased again after birth year 2000. Average single stranded DNA and double stranded DNA in Meruvax II were 142.05 ng/vial and 35.00 ng/vial, respectively, and 276.00 ng/vial and 35.74 ng/vial in Havrix respectively. The size of the fetal DNA fragments in Meruvax II was approximately 215 base pairs. There was spontaneous cellular and nuclear DNA uptake in HFF1 and NCCIT cells. Genes that have been linked to autism (autism associated genes; AAGs) have a more concentrated susceptibility for insults to genomic stability in comparison to the group of all genes contained within the human genome. Of the X chromosome AAGs, 15 of 19 have double strand break motifs less than 100 kilobases away from the center of a meiotic recombination hotspot located within an exon.



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HEALTH • MEDICINE

Whistleblower Claims CDC Covered Up Data Showing Vaccine-Autism Link

5 MINUTE READ

BY ALICE PARK ✕

AUGUST 28, 2014 5:13 PM EDT

If you haven't noticed, there's a war going on between those who believe in the health benefits of vaccines – that they can prevent deadly infectious diseases such as measles and polio – and those that believe that the immunizations do more harm than good. Now one of the authors of a 2004 government study that found similar vaccination rates among children with and without autism says the study omitted some important data.

The vaccine war is being fought on social media, in social circles and increasingly in doctor's offices, as physicians are faced with doubts and questions from parents who find themselves being recruited onto the side of skepticism. Skepticism is healthy, and the sign of curious minds, but not when it flies in the face of evidence. Especially gold standard, rigorous scientific

Dr Thompson said: 'I have waited a long time to tell my story and I want to tell it truthfully. I was involved in deceiving millions of taxpayers regarding the potential negative side effects of vaccines. We lied about the scientific findings.'

Dr Thompson said that the study, completed in 2001, showed that children receiving the MMR under age three were more likely to develop autism, particularly if they were male and African American. Results indicated that these boys were 2.64 times more likely to get an autism diagnosis than if they received the MMR after three years of age. American children, generally get their MMR shots at 12 and 18 months, but it was clear from the results that that was the most dangerous time. Instead of alerting the public, Dr Thompson claims the CDC destroyed the data, did not announce the results, embarking instead on a series of closed door meetings to discuss how best to present them.

It took four years before the CDC was ready to make any kind of announcement and when they did in 2004, it was that the MMR was safe.

Dr Thompson explained to him that in order to conceal the effect of the MMR, what they did was reduce the number of children in the study in order to reduce what is referred to as the statistical power – the ability of the study to detect the difference if one genuinely exists.

In the analysis plan, the rules which govern how the study will be conducted and which are set in stone, they had agreed to use two sources of information, the first was the children's school records and the second was the children's Georgia birth certificate record. While every child in the study had a school record, only half of the children in the study had a Georgia birth certificate record, because the other half had been born in other states.

The analysis plan was explicit, information on a child's race was to come from their school record, but when confronted by data that revealed an increased risk of autism in African American children, they deviated from their analysis plan. They chose to get the race data not from the school record but from the Georgia birth certificate record.

Dr Hooker said: 'Instead of having 3,000 individuals in the study, it went down to about 1,800. The relative risk went down from 2.64 to 1.8, but more importantly that relative risk was no longer statistically significant.'

Dr Thompson, who was in charge of the numbers, said: 'When we discussed the findings in these meetings, their opinion was that it couldn't be right. But the figures don't lie.'

Dr Thompson made his confession in many recorded conversations with former bioengineer Dr Brian Hooker, now associate professor at Simpson University. He didn't know he was being recorded.

Dr Hooker had been after the data for years as his own son Stephen, born in 1998, developed autism two weeks after getting his shots, including the MMR, age 15-months. He said: 'This was a time where CDC was just starting to do these studies on vaccines and autism. As a scientist I have over 60 technical and scientific publications in major international scientific journals and I was contacting the CDC and was deeply critical of their studies.'

'The CDC decided that the scientist who was going to interface with me at that time was Dr William Thompson. Because I was on his back, because they didn't like what I was saying about the statistics I received a letter from a CDC attorney in 2004 saying that I was no longer permitted to contact the CDC.'

In 2014, he got a phone call out of the blue from Dr Thompson, who said he wanted to tell all.

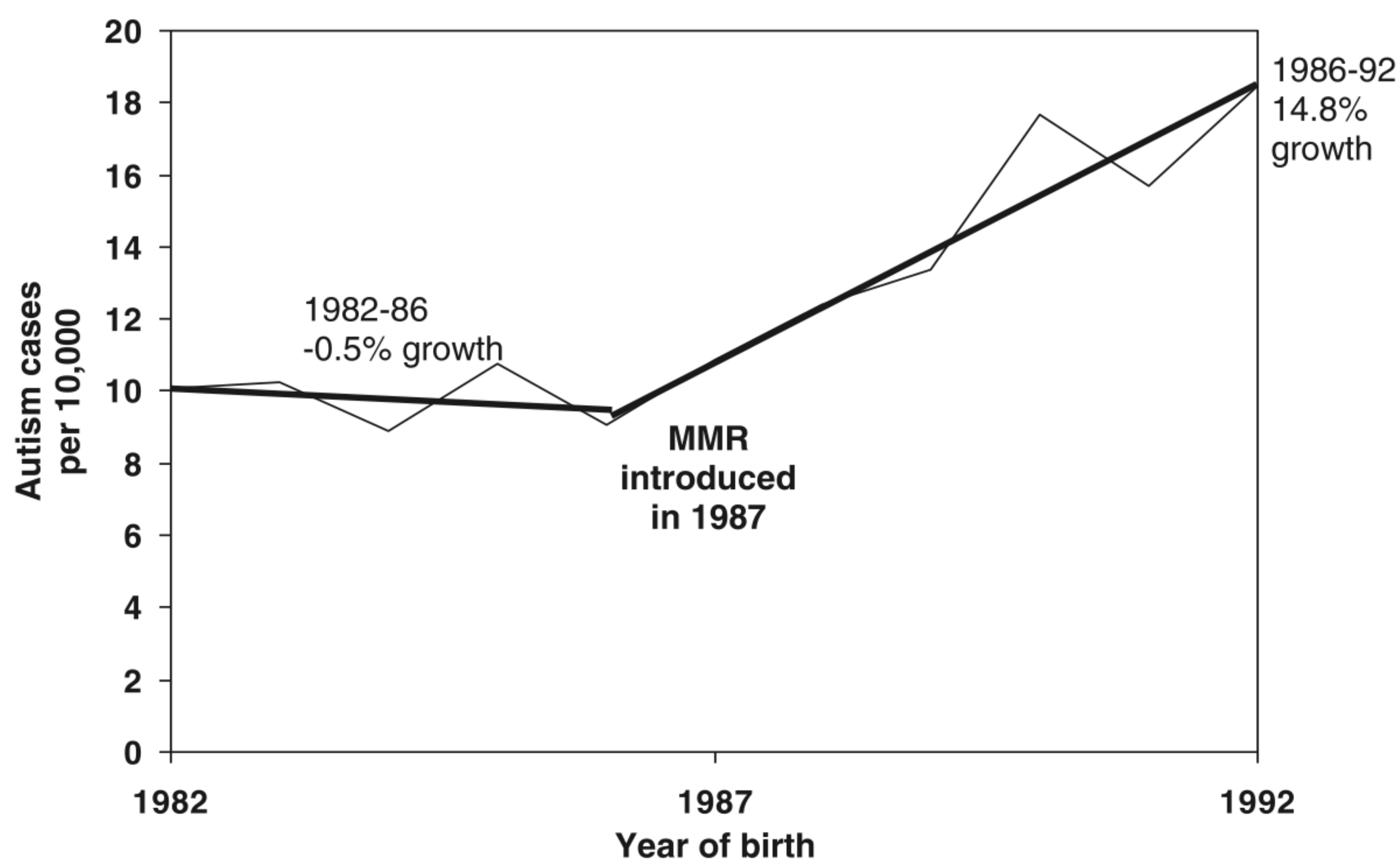


Figure 2. Autism prevalence in Denmark by year of birth, 1982-1992. Lines of best fit are shown for birth years 1982 to 1986, and from 1986 to 1992. Children born in 1986 were first to receive MMR in Denmark. The annual growth before MMR was -0.5% [trend = -0.15; 95% CI, (-1.06) - (-0.76), ns], compared with 14.8% after MMR introduction ($t = 6.94$, $p < 0.001$; trend 1.54, 95% CI, 0.97 - 2.11).

Source: Danish Psychiatric Central Register Data, with gratitude to Safe Minds.

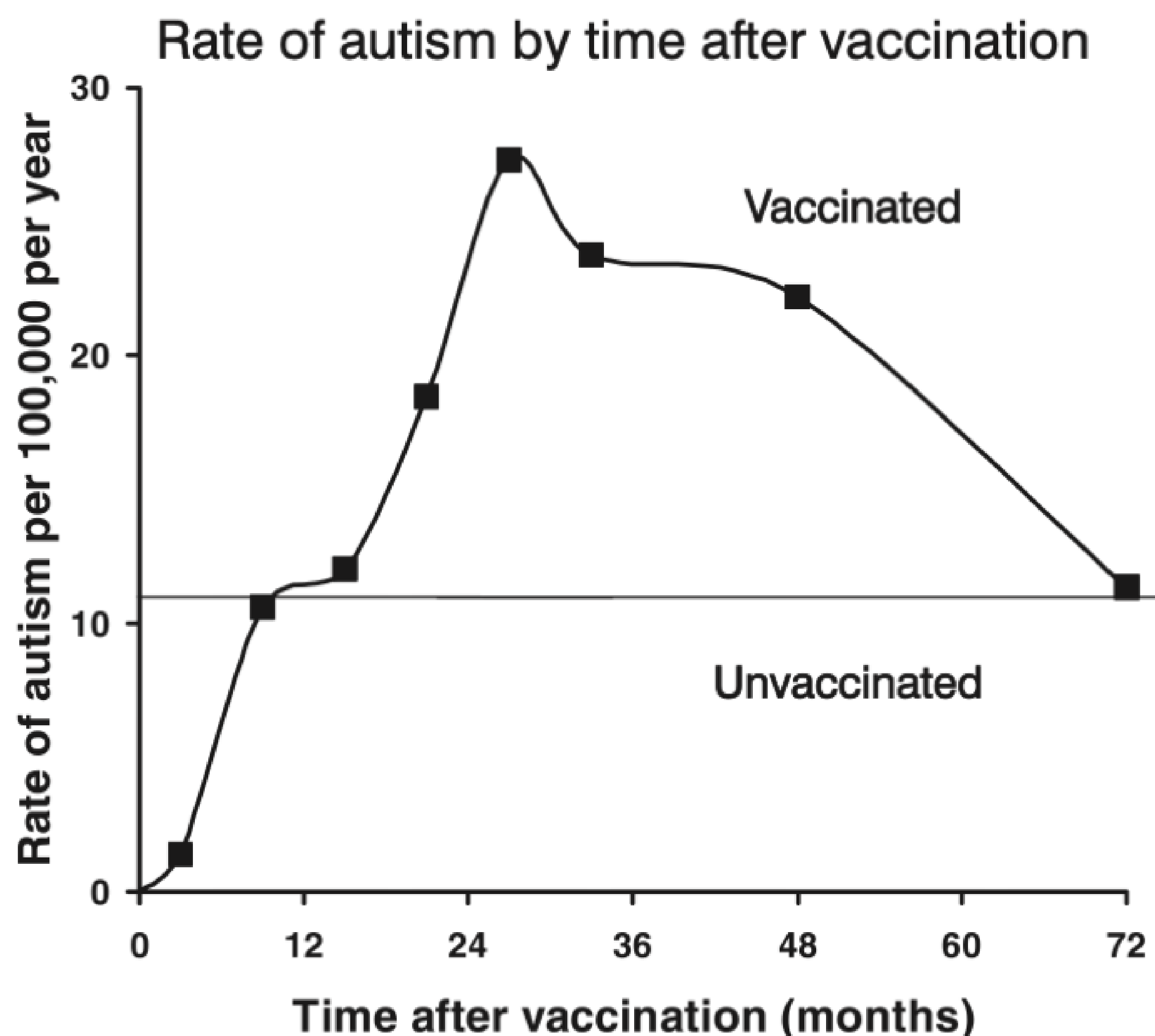


Figure 1. Incidence rate of autism in Denmark per 10^5 population per annum by time after vaccination vs. overall rate of 11.0 per 10^5 for the no- vaccination group.
Source: S. Suissa, Department of Epidemiology and Statistics, McGill University.

per 100,000 per year respectively for ages <15, 15-19, 20-24, 25-35 and >35 months. These rates are all above the overall rate of 11.0 for the reference group of no vaccination, over all ages. It is then somewhat implausible for the adjusted rate ratio to fall below 1, unless the risk profile by age in the unvaccinated is vastly different than in the vaccinated (effect-modification). In this case, the adjustment for age could have been artificial. It would be useful then to present rates on subjects 24-29 months since vaccination and on the unvaccinated (crude rate ratio 2.5) stratified by age. Otherwise, one could be tempted to conclude that the figure is in fact suggestive of an association between MMR vaccination and the risk of autism.

If, as is suggested, Madsen et al. adjusted inappropriately for age, then their findings need to be reinterpreted. In the absence of such adjustment, there is a statistically significant 45% excess risk of autism in recipients of the MMR vaccine and therefore, an association between MMR and autism in this Danish population.

Brief Report

FREE

March 7, 2001

Time Trends in Autism and in MMR Immunization Coverage in California

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[» Author Affiliations](#)

JAMA. 2001;285(9):1183-1185. doi:10.1001/jama.285.9.1183

Abstract

Context Considerable concern has been generated in the lay and medical communities by a theory that increased measles-mumps-rubella (MMR) immunization among young children may be the cause of an apparent marked increase in autism occurrence.

Objective To determine if a correlation exists in secular trends of MMR immunization coverage among young children and autism occurrence.

Design, Setting, and Participants Retrospective analyses of MMR immunization coverage rates among children born in 1980-1994 who were enrolled in California kindergartens (survey samples of 600-1900 children each year) and whose school immunization records were reviewed to retrospectively determine the age at which they first received MMR immunization; and of autism caseloads among children born in these years who were diagnosed with autism and were enrolled in the California Department of Developmental Services regional service center system.

Main Outcome Measures Measles-mumps-rubella immunization coverage rates as of ages 17 months and 24 months and numbers of Department of Developmental Services system enrollees diagnosed with autism, grouped by year of birth.

Results [Essentially no correlation](#) was observed between the secular trend of early childhood MMR immunization rates in California and the secular trend in numbers of children with autism enrolled in California's regional service center system. For the 1980-1994 birth cohorts, a marked, sustained increase in autism case numbers was noted, from 44 cases per 100 000 live births in the 1980 cohort to 208 cases per 100 000 live births in the 1994 cohort (a 373% relative increase), but changes in early childhood MMR immunization coverage over the same time period were much smaller and of shorter duration. Immunization coverage by the age of 24 months increased from 72% to 82%, a relative increase of only 14%, over the same time period.

Conclusions These data do not suggest an association between MMR immunization among young children and an increase in autism occurrence.

Association of autistic spectrum disorder and the measles, mumps, and rubella vaccine: a systematic review of current epidemiological evidence

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Affiliations + expand

PMID: 12860782 DOI: [10.1001/archpedi.157.7.628](https://doi.org/10.1001/archpedi.157.7.628)

Abstract

Objective: To systematically review the evidence for and against the existence of an association between autistic spectrum disorder (ASD) and the measles, mumps, and rubella (MMR) vaccine.

Study design: We conducted a systematic review of the medical literature to identify all controlled epidemiological articles examining for an association between ASD and the MMR vaccine. We extracted data from the articles on the characteristics and objectives of the study as well as evidence of an association.

Results: Twelve articles met the inclusion criteria. One study found no difference in the rates of ASD and the MMR vaccine in children who were vaccinated and those who were not. Six studies examined for evidence of an increase in ASD associated with an increase in the MMR vaccine coverage, none of which showed evidence of an association. Four studies examined if a variant form of ASD was associated with the MMR vaccine, none of which showed evidence of an association. Eight studies attempted to determine if there was a temporal association between developing ASD and receiving the MMR vaccine. Of these, 1 study identified an increase in parental concern in the 6-month period following vaccination with MMR in one of its analyses. The results of all other studies showed no association between ASD and the MMR vaccine.

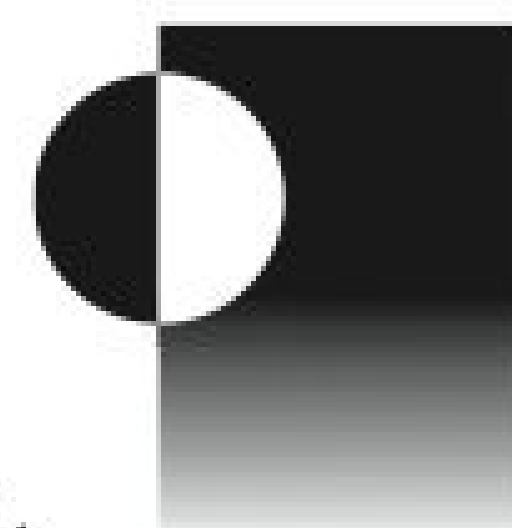
Conclusions: The current literature does not suggest an association between ASD and the MMR vaccine; however, limited epidemiological evidence exists to rule out a link between a rare variant form of ASD and the MMR vaccine.

AHFS Category: 80:08

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DTaP

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DESCRIPTION

Tripedia®, Diphtheria and Tetanus Toxoids and Acellular Pertussis Vaccine Adsorbed (DTaP), for intramuscular use, is a sterile preparation of diphtheria and tetanus toxoids adsorbed, with acellular pertussis vaccine in an isotonic sodium chloride solution containing sodium phosphate to control pH. After shaking, the vaccine is a homogeneous white suspension. Tripedia vaccine is distributed by Sanofi Pasteur Inc.

Additional Adverse Reactions:

- As with other aluminum-containing vaccines, a nodule may be palpable at the injection sites for several weeks. Sterile abscess formation at the site of injection has been reported.^{3,36}
- Rarely, an anaphylactic reaction (ie, hives, swelling of the mouth, difficulty breathing, hypotension, or shock) has been reported after receiving preparations containing diphtheria, tetanus, and/or pertussis antigens.³
- Arthus-type hypersensitivity reactions, characterized by severe local reactions (generally starting 2-8 hours after an injection), may follow receipt of tetanus toxoid.
- A few cases of peripheral mononeuropathy and of cranial mononeuropathy have been reported following tetanus toxoid administration, although available evidence is inadequate to accept or reject a causal relation.³⁷
- A review by the Institute of Medicine (IOM) found evidence for a causal relationship between tetanus toxoid and both brachial neuritis and Guillain-Barré syndrome.³⁷
- A few cases of demyelinating diseases of the CNS have been reported following some tetanus toxoid-containing vaccines or tetanus and diphtheria toxoid-containing vaccines, although the IOM concluded that the evidence was inadequate to accept or reject a causal relationship.³⁷

Adverse events reported during post-approval use of Tripedia vaccine include idiopathic thrombocytopenic purpura, SIDS, anaphylactic reaction, cellulitis, autism, convulsion/grand mal convulsion, encephalopathy, hypotonia, neuropathy, somnolence and apnea. Events were included in this list because of the seriousness or frequency of reporting. Because these events are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequencies or to establish a causal relationship to components of Tripedia vaccine.²

Reporting of Adverse Events

The National Vaccine Injury Compensation Program, established by the National Childhood Vaccine Injury Act of 1986, requires physicians and other health-care providers who administer vaccines to maintain permanent vaccination records of the manufacturer and lot number of the vaccine administered in the vaccine recipient's permanent medical record along with the date of administration of the vaccine and the name, address and title of the person administering the vaccine. The Act (or statute) further requires the health-care professional to report to the Secretary of the US Department of Health and Human Services, the occurrence following immunization of any events set forth in the statute or the Vaccine Injury Table, including